

Total No. of Questions : 9]

SEAT No. :

PE-895

[Total No. of Pages : 4

[6581]-1901

F.E.

ENGINEERING MATHEMATICS - I
(2019 Pattern) (Semester - I) (107001)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Q. 1 is compulsory.
- 2) Attempt Q.2 or Q.3, Q.4 or Q.5, Q.6 or Q.7, Q.8 or Q.9.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Figure to the right indicates full marks.
- 5) Use of electronic pocket calculator is allowed.
- 6) Assume suitable data, if necessary.

Q1) Write the correct option for the following multiple choice questions. **[10]**

a) If z is a homogeneous function of x and y and if $z = f(u)$ then $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ is

[1]

i) $\frac{f(u)}{f'(u)}$

ii) $n \frac{f(u)}{f'(u)}$

iii) $\frac{f'(u)}{f(u)}$

iv) $n \frac{f'(u)}{f(u)}$

b) The product of eigen values of a matrix A is equal to

[1]

i) Trace of A

ii) Determinant of A

iii) Rank of A

iv) None of the

c) The eigen values of $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ are

[2]

i) 0,0,0

ii) 0,0,1

iii) 0,0,3

iv) 1,1,1

- d) The rank of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ is [2]

- | | | | |
|------|---|-----|---|
| i) | 2 | ii) | 0 |
| iii) | 1 | iv) | 3 |

- e) If $u = x^2 + y^2$, $v = 2xy$ then $\frac{\partial(u,v)}{\partial(x,y)}$ is [2]

- i) $x^2 - y^2$ ii) $2(x^2 - y^2)$
iii) $4(x^2 + y^2)$ iv) $4(x^2 - y^2)$

- f) If $x = r \cos \theta$ and $y = r \sin \theta$ then $\left(\frac{\partial y}{\partial r}\right)_x$ is [2]

- i) $\frac{r^2}{y}$ ii) $\frac{r}{y^2}$

- $$\text{iii) } \frac{r^2}{y^2} \qquad \text{iv) } \frac{r}{y}$$

- Q2) a)** If $v = \frac{c}{\sqrt{t}} e^{(-x^2/4a^2t)}$ then show that $\frac{\partial v}{\partial t} = a^2 \frac{\partial^2 v}{\partial x^2}$. [5]

- b) If $u = f\left(\frac{y}{x}\right) + \sqrt{x^2 + y^2}$ prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = (x^2 + y^2)^{1/2}$. [5]

- c) If $u = f(x^2 - y^2, y^2 - z^2, z^2 - x^2)$ then find the value of $\frac{1}{x} f_x + \frac{1}{y} f_y + \frac{1}{z} f_z$. [5]

OR

- Q3) a)** If $x^2 = au + bv$, $y^2 = au - bv$ prove that $\left(\frac{\partial u}{\partial x}\right)_y \cdot \left(\frac{\partial x}{\partial u}\right)_v = \left(\frac{\partial v}{\partial y}\right)_x \cdot \left(\frac{\partial y}{\partial v}\right)_u$ [5]

- b) If $u = \sec^{-1}\left(\frac{x^3 + y^3}{x + y}\right)$ prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2 \cot u$. [5]

- c) If $z = f(x, y)$ and $x = e^u + e^{-v}$, $y = e^{-u} - e^v$ show that $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}$. [5]

- Q4) a)** If $x = u(1 - v)$, $y = uv$, prove that $JJ' = 1$. [5]
b) Prove that the functions $u = y + z$, $v = x + 2z^2$, $w = x - 4yz - 2y^2$ are functionally dependent and find the relation between them. [5]
c) Find the maximum and minimum values of $x^2 + y^2 + 8y + 15$. [5]

OR

- Q5) a)** If $u^3 + v^3 = x + y$, $u^2 + v^2 = x^3 + y^3$ Find $\frac{\partial(x, y)}{\partial(u, v)}$. [5]

- b)** A power dissipated in a resistor is given by $P = \frac{E^2}{R}$. Using calculus, find the approximate percentage error in P when E is increased by 3% and R is increased by 2%. [5]
c) Divide 24 into three parts such that the continued product of the first, square of the second and cube of the third may be maximum. [5]

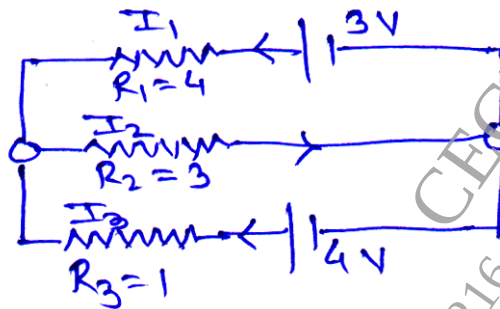
OR

- Q6) a)** Determine values of k for which the equations $2x - 3y + 5z = k$, $3x + y - z = 2$, $x + 4y - 6z = 1$ are consistent. [5]
b) Examine following set of vectors for linearly dependance. [5]
 $X_1 = (1, 2, 4)$, $X_2 = (2, -1, 3)$, $X_3 = (0, 1, 2)$, $X_4 = (-3, 7, 2)$.

- c)** Show that $A = \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{bmatrix}$ is an orthogonal matrix. [5]

OR

- Q7) a)** Solve the system of equations $x + y + 3z = 0$; $x - y + z = 0$; $-x + 2y = 0$; $x - y + 2z = 0$ [5]
b) Examine whether the set of vectors $X_1 = (5, 3, 2)$, $X_2 = (2, 1, 1)$, $X_3 = (-3, 1, 6)$ are linearly independent. [5]
c) Find the current I_1 , I_2 , I_3 in the circuit shown in the figures. [5]



Q8) a) Find eigen values and eigen vectors for the given matrix $A = \begin{bmatrix} 3 & 5 \\ -2 & -4 \end{bmatrix}$. [5]

b) Verify Cayley Hamilton theorem for following matrix $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. [5]

c) Find the modal matrix P which diagonalises the following matrix

$$A = \begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}. \quad [5]$$

OR

Q9) a) Find eigen values and eigen vector corresponding to largest eigen value

of a following matrix $A = \begin{bmatrix} 15 & 0 & -15 \\ -3 & 6 & 9 \\ 5 & 0 & -5 \end{bmatrix}$ [5]

b) Verify Cayley Hamilton theorem for following matrix and hence find A^{-1}

$$A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}. \quad [5]$$

c) Express the following quadratic form as "Sum of the squares form" by congruent transformation. Write down the corresponding linear transformation. [5]

$$Q(x) = 6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 + 4x_1x_3 - 2x_2x_3$$

